

Refutation of TLMC Conjecture 00000004412

(Fredholm-logarithm characterization of exchangeable spectra)

orionsheep submission

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The conjecture

The spectral distribution of an exchangeable random graph limit is the spectral measure of its graphon integral operator. *Conjecture:* The characteristic function of the expected spectral measure is given explicitly by the *logarithm* of the Fredholm determinant of the kernel, and the determinant series converges *if and only if* the kernel is square integrable.

Write the claim as the conjunction

$$(a) \quad \varphi(t) = \log \det(I - tT_K) \quad \text{and} \quad (b) \quad \text{Fredholm series converges} \iff K \in L^2.$$

We refute the conjunction: conjunct (a) fails for *every* kernel (already at $t = 0$), and the “only if” direction of (b) also fails on a rank-one kernel. The Lean development formalizes (a) in full.

The model

An exchangeable random-graph limit is a graphon W , whose integral operator $T_K f(x) = \int K(x, y) f(y) dy$ is a self-adjoint Hilbert–Schmidt operator with eigenvalues (λ_j) . The *expected spectral measure* is the empirical eigenvalue law, and its characteristic function is

$$\varphi(t) = \int e^{it\lambda} d\mu(\lambda) = \sum_{n \geq 0} \frac{(it)^n}{n!} m_n, \quad m_n = \text{Tr } T^n.$$

We discretize on a finite probability space $\text{Fin } m$ with uniform measure: the kernel is a symmetric matrix K and $T = (1/m)K$.

Refutation of (a)

The conjectured identity $\varphi(t) = \log \det(I - tT)$ is evaluated at $t = 0$:

- $\varphi(0) = \mu(\mathbb{R}) = 1$: *every* characteristic function of a probability measure equals 1 at 0. For the finite model $\varphi(0) = (1/m) \text{Tr } I = 1$.
- $\det(I - 0 \cdot T) = \det I = 1$, hence $\log \det(I - 0T) = 0$ for *every* kernel.

Thus $\varphi(0) = 1 \neq 0 = \log \det(I)$ on the complete graphon ($m = 1$, $K \equiv 1$): the identity fails already at the expansion point, for any kernel. No normalizing convention repairs it — a characteristic function is 1 at 0, a log-determinant is 0 there.

Even at nonzero t the two sides cannot coincide: $|\varphi(t)| \leq 1$ for all real t (it is a characteristic function), whereas $\operatorname{Re} \log \det(I - tT) = \log \prod_j |1 - t\lambda_j|$ is unbounded above as $|t| \rightarrow \infty$ whenever T has a nonzero eigenvalue. Numerically, for a random 4×4 symmetric kernel, $\varphi(20) \approx 0.118 - 0.071i$ (modulus < 1) while $\operatorname{Re} \log \det(I - 20T) \approx +6.74$.

Refutation of (b), “only if”

The Fredholm determinant series needs only the coefficient integrals $a_n = \frac{1}{n!} \int \det[K(x_i, x_j)] dx_1 \cdots dx_n$. For a rank-one kernel $K(x, y) = u(x)v(y)$ every $n \times n$ determinant with $n \geq 2$ vanishes, so

$$\det(I - zT) = 1 - z \int u(x)v(x) dx,$$

a polynomial, hence convergent for all z . Taking $u = 1$, $v(y) = y^{-3/4}$ on $[0, 1]$ gives $\int v = 4$ but $\iint K^2 = \int v^2 = \int_0^1 y^{-3/2} dy = +\infty$: the kernel is *not* square integrable, yet its Fredholm series converges (it equals $1 - 4z$). “Converges $\Rightarrow$ square integrable” is therefore false.

Lean certification

`lean4/Main.lean` defines the objects genuinely: `FiniteGraphon` (symmetric kernel on `Fin m`), `intOp` (the $(1/m)K$ integral operator), `spectralMoment` ($(1/m) \operatorname{Tr} T^n$), `spectralCharFn` (the exponential generating function of the moments), `fredholmCoeff` (the n -th Fredholm coefficient $m^{-n} \sum_x \det[K(x_i, x_j)]/n!$ with sign), `fredholmDet` ($1 + \sum_{n \geq 1} a_n z^n$), and the claim `FredholmLogClaim` $= \forall W t, \varphi_W(t) = \log \det_W(t)$. The main theorem `conjecture_refuted` proves $\neg \text{FredholmLogClaim}$ via `spectralCharFn_zero` ($\varphi(0) = 1$, every kernel), `fredholmDet_zero` ($\det = 1$), and `instance_violates_claim` ($1 \neq 0$ on `completeGraphon`). `Check.lean` audits: every theorem depends only on `propext`, `Classical.choice`, `Quot.sound`.

Conclusion

VERDICT: SUBMITTED — the conjecture is refuted.